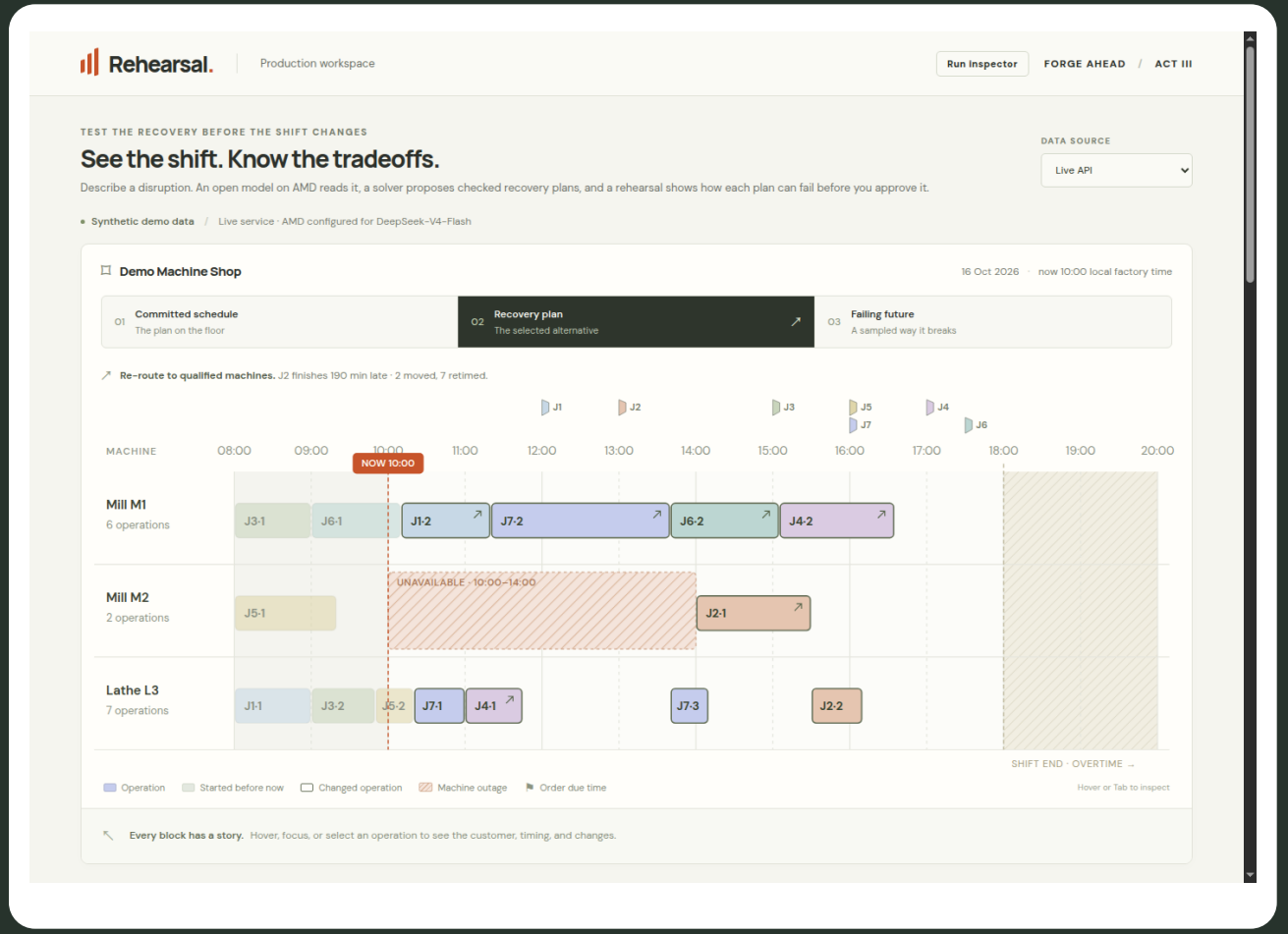
Rehearsal.

Test the recovery before changing the shift.

Checked recovery decisions for production planners.

Abhishek Sharma + Muskan Verma



Working AWS app • Real AMD extraction • Synthetic demo data

One breakdown.
Seven delivery promises.

The production planner has to decide what to move, what runs late, and what to approve.

10:00

M2 stops

Work is already in progress.

16:00

J7 must ship

Its hard deadline must hold.

A decision

Own the tradeoff

A revised plan needs a clear handoff.

FIRST USER A production planner in a make-to-order machine shop

Evidence / source

The note becomes a plan you can inspect.

> “Mill M2 is unavailable until 14:00. Order J7 still needs to ship at 16:00. Overtime requires approval.”

READ ON AMD | REVIEW | SOLVE + CHECK | REHEARSE | APPROVE + EXPORT

Machine outage

M2 · 10:00–14:00

Hard deadline

J7 · 16:00

Proposed option

Overtime · approval needed

Unknown machines and missing required times are flagged for human review.

Evidence / source

Save J7.
Show the cost.

KEEP ORIGINAL MACHINES

Infeasible

J7’s hard deadline cannot be met under this policy.

The hard date is not silently moved.

RE-ROUTE TO QUALIFIED MACHINES

J7 on time

J2: 190 minutes late

0 overtime minutes · Independent checker passed

That late delivery is an explicit concession for a person to approve.

Evidence / source

On paper, on time.
In rehearsal, exposed.

J4

239 / 500

simulated scenarios finish on time

Planned finish: 16:35 · Due: 17:00

500 samples · seed 7 · duration factors 0.9–1.3 · repairs 0–90 min late

Synthetic uncertainty; these counts are not measured factory reliability.

Evidence / source

A CONCRETE COUNTEREXAMPLE

Sampled future #24

J4 finishes at 18:01

61 minutes late. Its M1 operation waits 70 minutes longer than planned.

A decision someone owns. A record you can keep.

01

Accept the tradeoff

Required late-delivery and overtime concessions are explicit.

02

Approve a stored plan

The server checks its own schedule again before export.

03

Keep both exports

Dispatch CSV + a decision JSON with AMD trace and rehearsal evidence.

ACTUAL DISPATCH EXPORT · EXCERPT

```
machine job op start end
M1 J1 2 10:10 11:20
M1 J7 2 11:20 13:40
M1 J6 2 13:40 15:05
M1 J4 2 15:05 16:35
```

Approval locks the decision. Identical retries return the same exports.

Evidence / source

AMD reads the message.
The checks control the decision.

AMD TOKEN FACTORY

DeepSeek-V4-Flash

Typed proposals + exact source spans

WEB CONTAINER · CPU

React + FastAPI

OR-Tools CP-SAT → independent checker
→ NumPy rehearsal → approval / exports

VERIFIED LIVE REQUEST

Provider	**AMD Token Factory**
Model	**DeepSeek-V4-Flash**
Host	**developer.amd.com.cn**
Latency / attempts	**7,866 ms / 1 attempt**
Fallback	**Disabled; no fallback used**

No GPU replay or GPU speedup is claimed. The model API is the AMD workload.

Evidence / source

Small tests. Visible limits.

Extractor / labels	Complete changes	Wrong “OK”	Review raised
Rules / Muskan	**13 / 23**	2	9 / 12
AMD / Muskan	**23 / 23**	0	12 / 12
Same AMD / Abhishek	**21 / 23**	0	9 / 10

23 team-written messages; they informed prompt tuning. Two provisional label sets.
No independent benchmark or final reconciled extraction score is claimed.

Also recorded: 7 frozen synthetic recovery scenarios, including an infeasible case.
The repository retains unchanged outcomes and tradeoffs, not just improvements.

Evidence / source

A clear first user.
A measurable next step.

WHO USES IT

Production planners

Small, make-to-order machine shops handling disruption-driven replanning.

The aim: faster, clearer recovery decisions with an auditable handoff.

SHADOW PILOT

Use real historical disruptions

Replay cases alongside a planner's current workflow.

Measure the decision

Time to a correct approved plan; missed commitments and wrong accepted changes.

Start with the handoff

Validate CSV exports before connecting factory systems.

Current limits: synthetic data · one-day fixture · no ERP integration · no validated ROI

Evidence / source

Rehearse before you commit.

Abhishek Sharma + Muskan Verma

Intelligent Industry / AMD Developer Hackathon: ACT III

OPEN THE LIVE APP

VIEW CODE + EVIDENCE

github.com/AbhishekSharma55/amd-act-3

WATCH THE DEMO / youtu.be/XCLzMpVXXHU